作业

作业题目来自

梁昆淼《力学》 第 4 版 高等教育出版社

例： L01.08 梁昆淼书第 01 章第 08 题

Johns Analytical Mechanics 2nd ed Oxford University Press

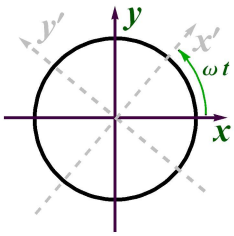
例： J02.05 Johns 书第 02 章第 05 题

J03.07† 带 † 表示仅要求荣誉班同学完成

Homework 06, Week 08 2023.10.23 Due 2023.10.30 L07.05, L07.09, J04.05, J04.10, J04.12

★ L07.05

质点在势力场中运动，力场具有轴对称。
以这对称轴作为静止坐标系 $O\,x\,y\,z$ 的 z 轴，
有取一以角速度 ω 绕对称轴旋转的转动
坐标系 $O\,x'\,y'\,z'$，写出质点的哈密顿函数 H'，
要求 H' 用质点相对于转动坐标系的坐标表出。
说明 H' 的物理意义，它是否为运动积分？



Solution

★ L07.09

质量为 m 的质点在引力场 $V=-\frac{k}{r}$ 中运动，假定一定的轨道平面为 $x\,y$ 平面，则系统的哈密顿函数为：$H=\frac{1}{2\,m}\left(p_x^2+p_y^2\right)-\frac{k}{r}$，

其中 $r=\sqrt{x^2+y^2}$. 角动量矢量沿 z 方向，其 z 分量为：$J=x\,p_y-y\,p_x$ 。龙格-伦兹矢量 $\vec{\boldsymbol{K}}$ 位于 $x\,y$ 平面内，其分量分别为：

$K_x=p_y\,J-\frac{m\,k\,x}{r}$，$K_y=-p_x\,J-\frac{m\,k\,y}{r}$ 试证：

(i) 角动量和龙格-伦兹矢量的两个分量与系统的哈密顿函数分别是对合的：$[J,\,H]=0$，$[K_x,\,H]=0$，$\left[K_y,\,H\right]=0$

即 J，K_x，K_y 为系统的运动积分；

(ii) 角动量和龙格-伦兹矢量的两个分量之间的泊松括号为：$[K_x,\,J]=-K_y$，$\left[K_y,\,J\right]=K_x$，$\left[K_x,\,K_y\right]=-(2\,m\,H)\,J$

如果限于考虑束缚态，即 $H=-|E|$，证明 $K_x\Big/\sqrt{2\,m\,|E|}$，$K_y\Big/\sqrt{2\,m\,|E|}$，J 与三维角动量的分量有相同的泊松括号

(iii) $K_x\Big/\sqrt{2\,m\,|E|}$，$K_y\Big/\sqrt{2\,m\,|E|}$，J 的平方和等于 $m\,k^2\big/(2\,|E|)$

Solution

一些 Mathematica 代码，通过反复代换，计算泊松括号。

★ J04.05

A system with two degrees of freedom has a Lagrangian : $L(q,\,\dot{q},\,t)=a\,\dot{q}_1^2+2\,b\,\dot{q}_1\,\dot{q}_2+c\,\dot{q}_2^2+f$ where a, b, c are given functions of q_1, q_2 and t

(a) Find the two generalized momentum $p_k(q,\,\dot{q},\,t)$ and write expressions for the $\dot{q}_k$ as functions of q, p, t

(b) Write the generalized energy function $H(q,\,\dot{q},\,t)$ and express it in terms of the proper phase space coordinates q, p, t to form the Hamiltonian $H\,(q,\,p,\,t)$

(c) Verify that the Hamilton equations for $\dot{q}_k$ simply restate the results from part (a).

(d) What do you learn from these results regarding to "declaration of independence" for the p_k variables?

Solution

★ J04.10

Consider a system consisting of a single particle.

(a) Using the phase space variables $q = x,\ y,\ z$ and $p = p_x,\ p_y,\ p_z$ prove that for any phase space function $f(q,\ p)$,

$$[f(p,\ q),\ p_x] = \frac{\partial f}{\partial x},\quad [f(p,\ q),\ p_y] = \frac{\partial f}{\partial y},\quad [f(p,\ q),\ p_z] = \frac{\partial f}{\partial z}.$$

(b) The orbital angular momentum of a single mass is $\vec{\boldsymbol{L}} = \vec{\boldsymbol{r}} \times \vec{\boldsymbol{p}}$. Prove that

$$[L_z,\ x] = y,\quad [L_z,\ y] = -x,\quad [L_z,\ z] = 0 \ \text{ and } \ [L_z,\ p_x] = p_y,\quad [L_z,\ p_y] = -p_x,\quad [L_z,\ p_z] = 0$$

Solution

★ J04.12

(a) Veify that the transformation $\begin{cases} Q_k = p_k \\ P_k = -q_k \end{cases}$ satisfy the following conditions and hence is canonical

$$[Q_k,\ Q_l] = 0,\quad [Q_k,\ P_l] = \delta_{k\,l},\quad [P_k,\ P_l] = 0\ .$$

(b) Consider the Hamiltonian in $H(q,\ p,\ t) = \dfrac{p_x^2 + p_y^2 + p_z^2}{2\,m} + U(x,\ y,\ z,\ t)$.

Use the canonical transformation above to write $H\,(P,\ Q,\ t)$

(c) Write the Hamiltonian equations in both the $q,\ p$ and $Q,\ P$ systems, and show that they are equivalent,

in the sense that each set of equations implies the other set.

Solution